

Carl Andrew Ziegler

Exoplanet Astronomer | Associate Professor | Director, SFA Observatory

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Observational astrophysicist and university educator leading international high-resolution imaging programs for Kepler and TESS. Designs survey measurements and automated Python pipelines, investigates exoplanets in multiple-star systems, and translates research into open-source software, authentic student research, and public astronomy programs.

180+

refereed publications

5,400+

career citations

49

Scholar h-index

4,800+

Kepler/TESS hosts

20+

researchers mentored

Expertise

Research

Exoplanet characterization; stellar multiplicity; speckle interferometry and adaptive optics; survey design; selection effects; Bayesian inference; Monte Carlo methods; time-series photometry.

Technical

Python, Astropy, emcee, JavaScript, HTML/CSS, C++, LabVIEW, Mathematica, Zemax, and SolidWorks; automated reduction pipelines, PSF subtraction, astrometry, contrast curves, robotic telescope control, and scientific visualization.

Education

Ph.D., Physics, 2018

University of North Carolina at Chapel Hill

Dissertation: Characterization of Exoplanets and Stellar Systems with New Robots

M.S., Physics, 2013

Southern Illinois University Carbondale

B.A., Physics and Mathematics, 2009

William Jewell College

Selected Publications

- **Ziegler, C.**, et al. (2026), "SOAR TESS Survey III: Mapping the Stellar Companion Deficit around TESS Planet Candidates." [arXiv:2608.26250](https://arxiv.org/abs/2608.26250); submitted to *AJ*.
- **Ziegler, C.**, et al. (2021), "SOAR TESS Survey II: The Impact of Stellar Companions on Planetary Populations." *AJ*, 162, 192.
- **Ziegler, C.**, et al. (2020), "SOAR TESS Survey I: Sculpting of TESS Planetary Systems by Stellar Companions." *AJ*, 159, 19.
- **Ziegler, C.**, et al. (2018), "Measuring the Recoverability of Close Binaries in Gaia DR2 with Robo-AO." *AJ*, 156, 259.

Selected Presentations

2025 Invited Colloquium

"Double Suns, Lonely Worlds: How Binary Stars Shape the Hunt for Exoplanets."

East Texas A&M University, Department of Physics and Astronomy.

2024 Oral Presentation

"SOAR TESS Survey: Thousands of High-Resolution Images of Planet-Hosting Stars."

Joint Texas Sections APS/AAPT and Zone 13 SPS Meeting.

Academic Appointments

Associate Professor of Astronomy with Tenure

Stephen F. Austin State University

2020–Present

Nacogdoches, TX

- Direct the SFA Observatory and coordinate observing programs, student training, public events, and regional partnerships.
- Teach across introductory astronomy, astrophysics, observational astronomy, mathematical methods, and senior practicum; design authentic assessments and research experiences.
- Mentored more than 20 undergraduate and graduate researchers, producing refereed coauthorships, conference presentations, REU placements, and graduate-school and technical-career outcomes.

Dunlap Postdoctoral Fellow

Dunlap Institute, University of Toronto

2018–2020

Toronto, Canada

- Founded the SOAR TESS Survey and One-Hit Wonders program; developed automated target-selection, observing, reduction, and validation workflows.

Research Leadership

SOAR TESS Survey, Principal Investigator (2018–Present). Lead a systematic HRCam speckle-imaging program on the 4.1-m SOAR Telescope. The cumulative census covers 2,982 targets and 731 source detections around 689 hosts. Paper III finds 3.5 times fewer companions inside 50 au and 2.6 times fewer inside 100 au than the field-star expectation. Coordinate target selection, telescope operations, collaborators, student researchers, data products, and publications.

Robo-AO Kepler Survey and Robo-AO 2. Collaborators and I imaged all 3,857 Kepler planet-candidate systems, supporting validation of more than 1,500 planets. Co-I on NASA's selected TESS Cycle 9 GI program, with **\$162,000 in Phase 2 funding** to continue high-resolution TESS follow-up using Robo-AO 2 through a dual-band visible/near-infrared survey of about 4,000 TOIs.

Gravitas, Creator and Lead Developer (2025–Present). Designed and deployed an open-source browser-based astronomy simulation for orbital dynamics, black holes, energy conservation, spacetime, gravitational-wave sonification, and exoplanet transit light curves; co-developed it with an undergraduate researcher and integrated it into guided-inquiry instruction.

Teaching, Mentorship, and Outreach

- **Documented teaching effectiveness:** 2025 annual student-evaluation average of 4.85/5; evaluations regularly exceed 4.5/5. Formal peer and administrative reviews document sustained growth and pedagogical innovation.
- Rebuilt Observational Astronomy as a semester-long authentic research experience and developed an upper-level Astrophysics II course with original notes, problem sets, examinations, and worked solutions; completed four ACUE courses and SFA Certified Online Instructor training.
- Organized more than 20 public observing nights since 2021, reaching more than 2,000 visitors; led a 2023 annular-eclipse event for more than 200 attendees and supported total-eclipse programming, school visits, Scout events, camps, and regional STEM partnerships.

Professional Service

NASA and journal proposal reviewer; referee for *MNRAS*, *AJ*, *ApJ*, *PASP*, and *A&A*; university and college committee service; public astronomy programming and regional STEM curriculum development.